DATA SCIENCE & ANALYTICS INTERNSHIP

Task 3 : Marketing Funnel & Conversion Performance Analysis

Submitted by : Kripa K

Internship : Future Interns

Domain : Data Science & Analytics

1. Introduction

Marketing Funnel and Conversion Performance Analysis is a data analytics project that focuses on understanding customer behavior on an e-commerce platform. It examines how users interact with products, add items to their carts, and complete purchases.
The main objective of this project is to analyze customer interactions, identify potential drop-off points, understand product purchasing patterns, and recommend improvements to the shopping experience.
Python and Power BI were used to analyze the dataset and present the results through visualizations and key performance indicators.

Dataset Description
- Dataset Name: E-commerce Behavior Data from October 2019
- Source: Kaggle
- Number of Records: 99,983
- Number of Columns: 9
- Purpose: To analyze customer interactions, product views, cart additions, purchases, and purchasing patterns.

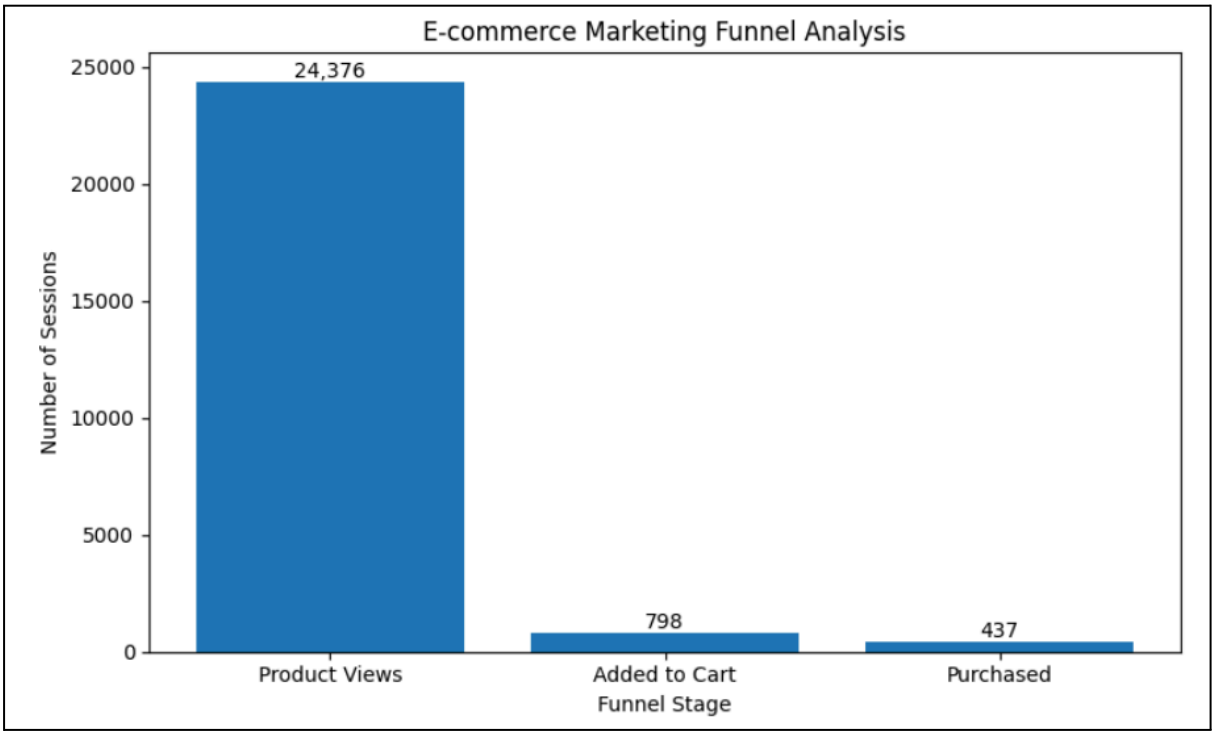
Problem Statement
E-commerce businesses need to understand customer shopping behavior and identify areas where potential purchases may be lost.
Without proper analysis, businesses may struggle to determine why customers browse products without adding them to their carts or completing purchases.
This project analyzes e-commerce event data to examine customer interactions, identify potential funnel drop-offs, compare product category performance, and provide recommendations to improve the shopping experience.

Project Overview
The project was carried out using Python, Pandas, Matplotlib, Jupyter Notebook, and Power BI.
First, the dataset was loaded, inspected, and prepared for analysis. A sample containing 99,983 records was used for the project.
Next, customer events, session-level funnel progression, product categories, daily purchase activity, and average product prices were analyzed.
Finally, a Power BI dashboard was created to present the key performance indicators and visual findings in an easy-to-understand format.

2. Data analysis and Visualization

2.1 E-commerce Marketing Funnel Analysis

Graph : E-commerce Marketing Funnel Analysis

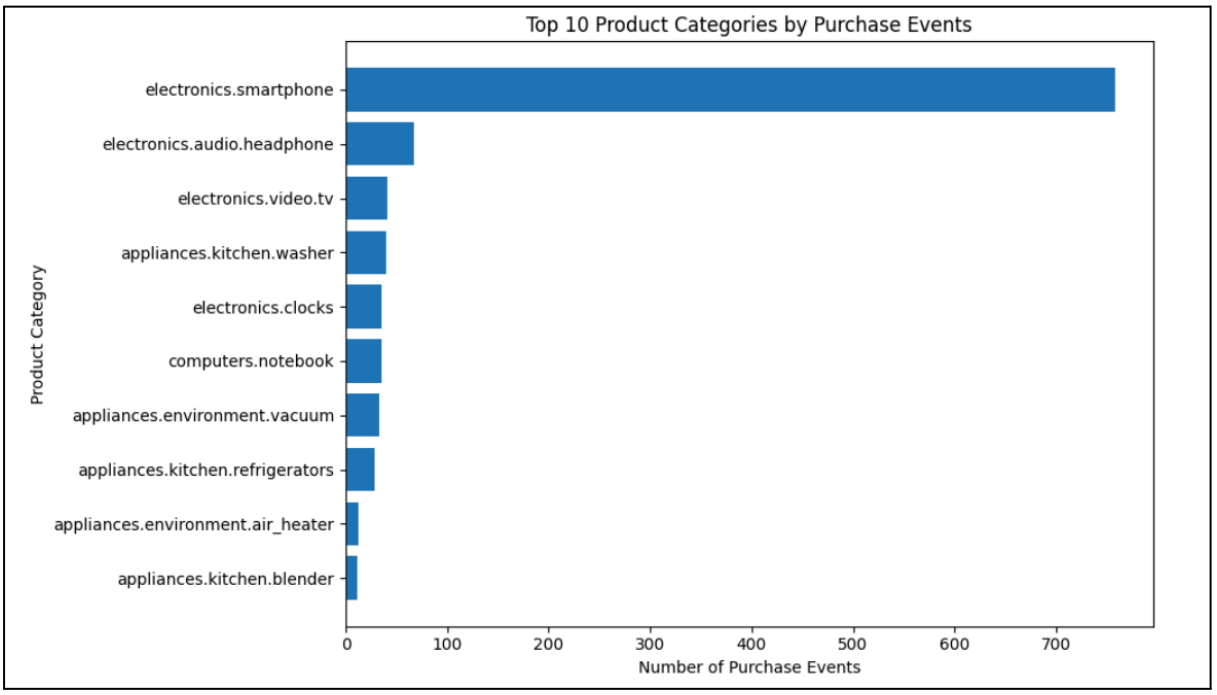


Interpretation:
The marketing funnel analysis shows how many user sessions progress through different stages of the e-commerce shopping journey. The analysis identified 24,376 sessions with product views, 798 sessions with a cart addition following a view, and 437 sessions with a subsequent purchase. The view-to-cart rate is approximately 3.27%, while the cart-to-purchase rate is approximately 54.76%.

Business Insight:
The largest drop-off occurs between product viewing and adding products to the cart. Businesses should improve product descriptions, images, pricing transparency, and the visibility of add-to-cart buttons to encourage customers to proceed to the next stage. Simplifying the shopping experience may help improve conversion performance.

2.2 Top 10 Product Categories by Purchase Event

Graph : Top 10 Product Categories by Purchase Event

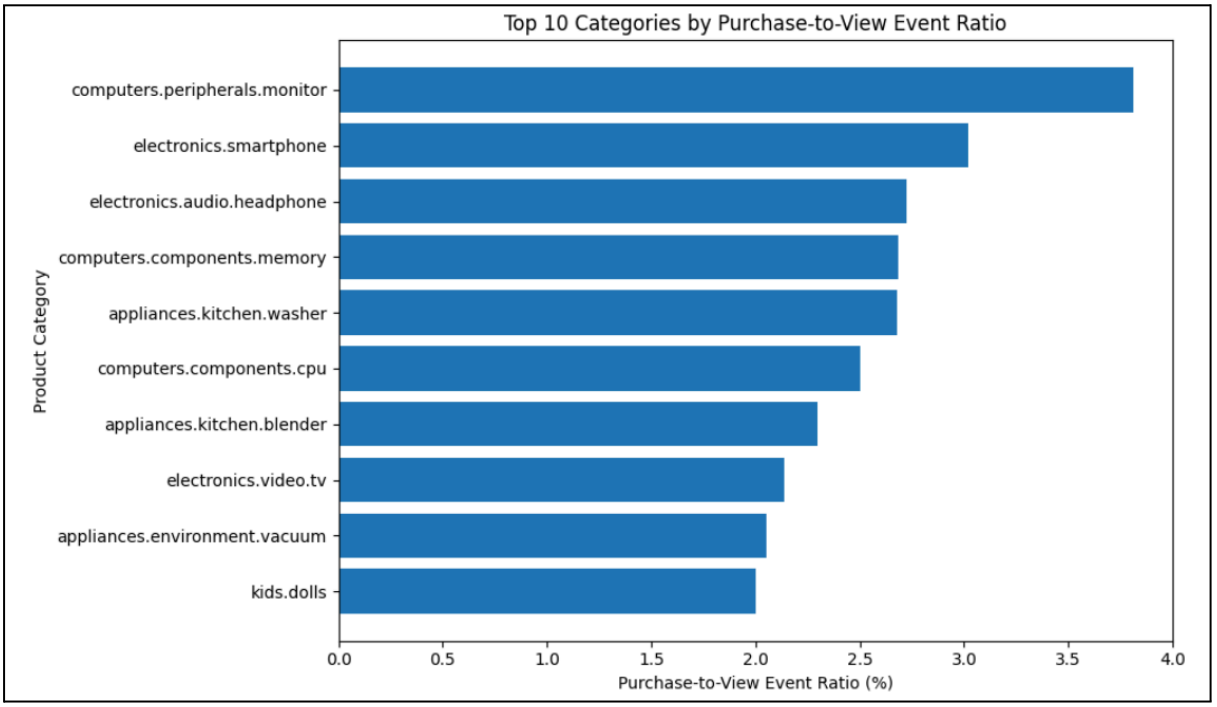


Interpretation:
This graph displays the ten product categories with the highest numbers of recorded purchase events in the analyzed dataset. It helps compare customer purchasing activity across different product categories and identify categories that contribute more to recorded purchases.

Business Insight:
Businesses can use the results to identify categories that may deserve greater attention in inventory planning, product recommendations, and promotional campaigns. Categories with lower purchase counts can also be investigated to understand whether product visibility, availability, or customer demand needs improvement.

2.3 Top 10 Categories by Purchase-to-View Event Ratio

Graph: Top 10 Categories by Purchase-to-View Event Ratio

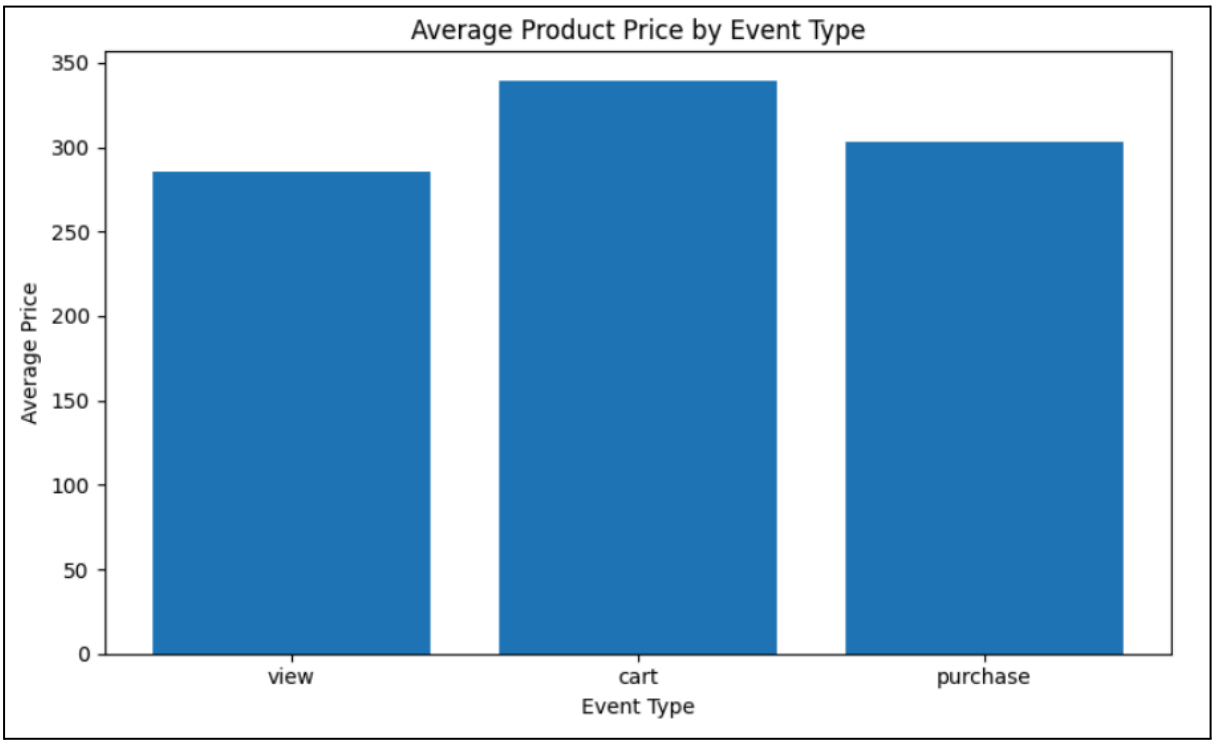


Interpretation:
This graph compares product categories based on the ratio of purchase events to view events. Categories with higher ratios have relatively more recorded purchase events compared with their view events. The analysis excludes categories with fewer than 100 view events to reduce the influence of very small counts.

Business Insight:
Businesses can investigate categories with higher purchase-to-view ratios to understand which products may be attracting stronger buying interest. Similar product recommendations, improved product visibility, and targeted promotions may help other categories perform better.

2.4 Average Product Price by Event Type

Graph: Average Product Price by Event Type



Interpretation:
The graph compares average product prices associated with view, cart, and purchase events. The analysis shows average prices of 285.88 for views, 339.66 for cart additions, and 302.83 for purchases. The average price associated with cart events is higher than the average price associated with purchase events.

Business Insight:
Businesses can examine purchasing patterns across different price ranges to understand how product prices relate to customer activity. They may test suitable promotions, improve product recommendations, and review pricing strategies. However, these average prices alone cannot establish that price causes customers to abandon their carts.

3. Key Business Insights

The analysis of e-commerce behavior data revealed several useful findings about customer interactions and purchasing patterns.

3.1 High Product Browsing Activity

The dataset contains 97,121 product view events, indicating substantial product browsing activity in the analyzed sample.
Insight: Product browsing provides an opportunity to encourage customers to consider purchasing products.

3.2 Low View-to-Cart Progression

Only 798 sessions contained a cart addition after a product view, representing approximately 3.27% of sessions with product views.
Insight: The transition from browsing to adding products to the cart is an important area for further investigation.

3.3 Purchase Activity After Cart Addition

A total of 437 sessions contained a product view followed by a cart addition and then a purchase.
Insight: Businesses should examine both the factors that encourage purchase completion and the reasons some shopping sessions do not progress through the funnel.

3.4 Differences in Product Category Performance

The category analysis identifies the ten categories with the highest recorded purchase counts.
Insight: Category-level purchasing patterns can support product promotion and inventory planning decisions.

3.5 Differences in Average Product Prices

The average price associated with cart events was 339.66, compared with 302.83 for purchase events.
Insight: Further investigation of price ranges and purchasing behavior may help businesses evaluate suitable pricing and promotional strategies.

3.6 Daily Purchase Trends

The daily purchase trend chart shows changes in recorded purchase activity over the analyzed period.
Insight: Monitoring daily purchasing patterns can help businesses understand changes in demand and plan future promotional activities.

3.7 Marketing Attribution Limitation

The selected dataset does not contain marketing channel or campaign information.
Insight: The analysis cannot establish which advertising channel generates the highest-quality leads. Additional marketing attribution data would be required to compare channel performance and campaign effectiveness.

Summary

The findings highlight opportunities to investigate the transition from product views to cart additions, understand product category performance, monitor purchase activity, and improve the shopping experience using data-driven decisions.

4. Business Recommendations

4.1 Improve Product Pages

Provide clear product descriptions, high-quality images, accurate prices, and complete product information.Customers may find it easier to evaluate products and make purchasing decisions.

4.2 Improve Add-to-Cart Visibility

Make add-to-cart buttons easy to locate and use on both desktop and mobile devices.A simpler product-page experience may encourage more customers to add products to their carts.

4.3 Simplify the Checkout Process

Reduce unnecessary checkout steps, confusing forms, and unexpected charges.A smoother checkout experience may help reduce purchase abandonment.

4.4 Review Pricing and Promotions

Analyze purchasing behavior across different price ranges and consider suitable discounts or promotional offers.Appropriate pricing strategies may encourage purchases while supporting business profitability.

4.5 Focus on High-Performing Categories

Use the top product category results to guide inventory planning, product recommendations, and category-specific promotions.Businesses can make more informed decisions about product availability and promotional priorities.

4.6 Monitor Daily Purchase Activity

Track purchase trends regularly and investigate significant changes in activity.Monitoring demand can help businesses plan promotional activities and respond to changing customer behavior.

4.7 Track Funnel Performance

Measure product views, cart additions, purchases, and clearly defined conversion rates consistently.Businesses can identify improvement areas and evaluate the effectiveness of changes over time.

4.8 Collect Marketing Channel Data

Connect customer events with marketing source, medium, and campaign information.This would enable future analysis of channel performance, campaign effectiveness, and marketing return on investment.

4.9 Test Improvements

Use A/B testing to compare different product page designs, calls to action, and checkout processes.Testing helps businesses determine which changes produce measurable improvements.

Summary

Implementing these recommendations and monitoring the results can help businesses improve the shopping experience, identify conversion opportunities, and make more informed decisions.

5. Power BI Dashboard (Tool Used : Microsoft Power BI Desktop)

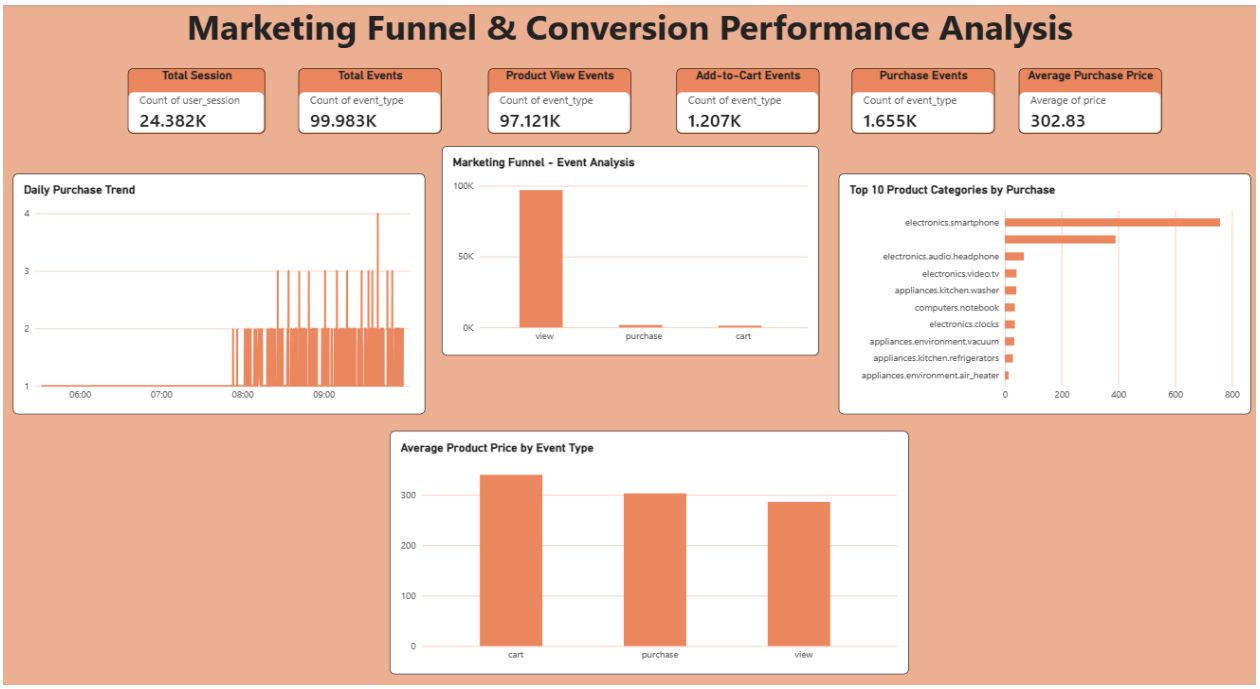
Dashboard Description

An interactive Power BI dashboard was developed to analyze e-commerce customer behavior, product views, cart additions, purchase events, and product pricing. The dashboard was created using a sample of approximately 100,000 event records from the E-commerce Behavior Data from Multi-Category Store dataset (October 2019). It presents key performance indicators (KPIs) and visualizations to make the results easier to understand.

Key Performance Indicators (KPIs)

The dashboard includes the following KPI cards:
- Total Sessions: 24,382
- Total Events: 99,983
- Product View Events: 97,121
- Add-to-Cart Events: 1,207
- Purchase Events: 1,655
- Average Purchase Price: 302.83

Figure : Marketing Funnel & Conversion Performance Analysis



Dashboard Visualizations

The dashboard contains the following visualizations:
1. Marketing Funnel – Event Analysis: Compares product views, cart additions, and purchase events.
2. Daily Purchase Trend: Displays purchase activity over time.
3. Top 10 Product Categories by Purchases: Identifies categories with the highest recorded purchase-event counts.
4. Average Product Price by Event Type: Compares average prices associated with views, cart additions, and purchases.

Key Dashboard Insights

The dashboard shows that product view events account for the largest share of recorded activity. Cart additions are considerably fewer than product views, highlighting an area for further investigation. The category comparison helps identify product categories with higher purchase activity, while the price comparison provides an overview of average prices across different event types.

Business Value

This dashboard helps summarize e-commerce activity and supports data-driven decisions related to product visibility, inventory planning, promotional strategies, and customer shopping experiences. It provides a clear overview of the analyzed sample and helps identify areas for further investigation.

Dataset Note: The dashboard uses approximately 100,000 event records rather than the complete original dataset. The dataset does not contain explicit marketing-channel or campaign information; therefore, channel-specific performance cannot be evaluated.

6. Conclusion

The Marketing Funnel and Conversion Performance Analysis project examined e-commerce behavior data to understand customer interactions, purchasing patterns, and potential opportunities for improving the shopping experience.

Python, Pandas, Matplotlib, Jupyter Notebook, and Power BI were used to prepare the data, perform analysis, create visualizations, and present the findings through a dashboard.

The analysis identified a substantial difference between product view events and cart addition events. The simplified session-level funnel also highlighted the progression of sessions from product views to cart additions and purchases. In addition, category-level purchasing patterns, daily purchase activity, and average product prices were examined.

The Power BI dashboard summarized the results using KPI cards and charts, making the findings easier to understand and communicate.

Based on the findings, recommendations were proposed to improve product pages, make add-to-cart options more visible, simplify checkout, review pricing strategies, monitor purchase trends, and track funnel performance consistently.

Although the selected dataset did not contain marketing channel or campaign information, it provided a useful foundation for analyzing e-commerce customer behavior and identifying potential areas for improvement.

Overall, this project strengthened practical skills in data cleaning, exploratory data analysis, data visualization, conversion analysis, KPI development, and dashboard creation. It demonstrated how data analytics can support business understanding and more informed decision-making.